



A fuzzy planar subgraph formation model for partitioning very large-scale integration networks

Shriram Kalathian^a, Sujatha Ramalingam^{b,*}, Nagarajan Deivanayagampillai^c

^a Department of Mathematics, St. Joseph's Institute of Technology, Chennai, 600019, Tamil nadu, India

^b Mathematics, School of Science and Humanities, Shiv Nadar University Chennai, Kalavakkam, 603110, Tamil nadu, India

^c Department of Mathematics, Rajalakshmi Institute of Technology, Chennai, 600124, Tamil nadu, India

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ABSTRACT

This study focuses on forming planar subgraphs in fuzzy graphs where the planarity value is not equal to 1. The paper commences by introducing the concepts of vertex-deletion (VD) and edge-deletion (ED) operations applied to fuzzy graphs. These operations aim to derive the fuzzy planar subgraphs and extensively examine the consequential significant findings. Additionally, maximal and maximum planar subgraphs in vertex-deletion (VD) and edge-deletion (ED) are presented alongside illustrative examples. A novel approach called the Planar Partition subgraph (PP subgraph) is introduced for identifying the fuzzy planar subgraphs from a fuzzy graph with an algorithm rooted in PP subgraphs for enhanced efficiency. Furthermore, the concept of thickness value in fuzzy graphs is constructed using PP -subgraphs is introduced. The discussion involves the thickness value of a fuzzy planar graph and its connection to the planarity value, supported by illustrative examples. Finally, an application made on PP subgraphs where the network needs to be partitioned into a finite number of subgraphs is demonstrated in a very large-scale integration network.

1. Introduction

“Subgraph” is one of the known terminology predominantly used in classical graph theory. Generally subgraphs provides a comprehensive information about the characteristics and properties of a graph. It has wide range of usages in various fields such as bioinformatics, chemoinformatics, social networks, traffic analysis, etc. Further applications of subgraph includes graph clustering, where frequent subgraphs are used to detect the communities in networks more effectively. Another important application is graph mining, which provides the detailed analysis of the network subgraphs and reveal the local structure of the network. Typically, a subgraph is obtained from a graph by spanning, deleting vertices and edges, and contracting edges.

Even fuzzy subgraph has an advanced extension of classical theory that includes fuzziness in the sub-connections. The membership values reflects as the strength of the connections between the pair of vertices. Clearly, fuzzy subgraphs are subsets of nodes and edges within the fuzzy graph and capturing a portion of the original fuzzy graph. For instance, consider the social network of a family is assumed to be fuzzy graph where the relationship between the members can be defined to be strong, intermediate and weak, provides a way to understand the nature of the interactions. The particular member network is considered to be fuzzy subgraph.

The paper designed as follows: In Section 2, previous works and methodology of fuzzy subgraph are shown, and the necessary and basic definitions with needed theorems are shown in Section 3. The concept of vertex-deletion, edge-deletion with maximum and maximal fuzzy planar subgraphs are provided in Sections 4 and 5 respectively. To obtain the planar from the fuzzy graph by Planar partition subgraph (PP -subgraph) and the thickness property is shown in Sections 6 and 7 respectively. Finally, Section 8 provides application of PP -Subgraphs in VLSI Networks particularly in ISCAS85 C-17 benchmark circuit (ISCAS85 C-17 referred as International Symposium on Circuits and Systems in 1985 Circuit 17) followed by conclusions and future works.

2. Literature review

2.1. Previous works

Fuzzy graph theory, an extension of classical graph theory [1,2], emerged as a powerful tool to represent and analyze the complex systems where uncertainty and imprecision are intrinsic. Fuzzy graphs incorporate the concept of fuzziness into graph structures, allowing for the representation and analysis of relationships that may not be well-defined. Over the years, researchers have explored various findings in

* Corresponding author.

E-mail addresses: shriram.bhavani5@gmail.com (S. Kalathian), sujathar@snuhennai.ac.in (S. Ramalingam), dnrmsu2002@yahoo.com (N. Deivanayagampillai).

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graphs as well as fuzzy graphs. In this study, our attention is directed towards planar graphs, Harary [3] and Wilson [4] unveiled intriguing traits in both planar, non-planar and dual graphs. This literature review provides an overview of key contributions and advancements in the field of fuzzy planar graphs.

The concept of fuzzy sets introduced by Zadeh (1965), which provides an element has a membership value. Rosenfeld's idea on fuzzy graphs have the edges and vertices are associated with degrees of membership value, which allows the vertices and edges have certain relationships. The pioneering work of Rosenfeld (1975) [5] formalized the notion of a fuzzy graph, defining its basic components and operations. In Modersen et al. [6,7], and Pal et al. [8] provides the basic results and the recent developments are portrayed with real-life applications. Additionally, the realm of fuzzy graph has numerous applications. This applications includes various domains such as social networks [9], radio frequency analysis [10], power control systems by m -polar fuzzy threshold graphs [11], various structures [12], human trafficking [13], different models in intuitionistic fuzzy graph [14,15], applications related to total fuzzy graphs [16], and the significance of connectivities [17]. Also, many interesting research works associates with decision analysis based on fuzzy are proposed and further [18–20] helps to understand the importance of the notions on the realms.

Abdul-Jabbar [21] marked a pivotal juncture to integrate the fuzzy graphs with planar graphs, which glows as new venture in the environment of fuzzy graphs. Samanta & Pal in their contributions, not only introduced the novel concepts of fuzzy planar graphs, also delved into the nature of fuzzy planarity values fortified by compelling theorems. Their insights were further validated through the image processing and segmentations in [22–24]. Complementing this, Pramanik [25] investigates into the interval-valued fuzzy graphs, enriching the discourse on planar graphs. Amidst these developments, emergent frontiers have emerged, encompassing Pythagorean fuzzy planar graphs [26,27], single-value neutrosophic planar graphs [28], generalized neutrosophic planar graphs [29], interval-valued m -polar fuzzy planar graphs [30] and m -polar interval valued intuitionistic fuzzy planar graphs [31], etc. Currently, the utilization of fuzzy planar graph methodologies is advancing to the higher stages like fractional and edge colorings [32,33]. These new territories collectively contribute to the evolving of fuzzy planar graph theory. This section shows the concept of fuzzy planar graphs extended to its different types with useful applications are portrayed. Now, this study distinguishes itself from the present scenario and follows a distinct trajectory from the this structures, commencing with an exploration of subgraphs and their planar characteristics.

Fuzzy graphs and classical graphs have many advantageous in decision-making [34,35] and exhibit intriguing real-world applications. This review delves into the substantial role played by fuzzy graphs in facilitating decision-making processes. It showcases their utility in diverse domains, such as crime detection in marine and road transport, as evidenced by the study in [36]. Furthermore, this review explores the application of multi-criteria group decision-making (MCGDM) techniques in selecting the optimal critical union accomplice for the selected bank under a spherical fuzzy environment, as elucidated in [37]. Another dimension of this examination encompasses the use of decision-making techniques involving picture fuzzy graphs, with a case study in the context of investment decisions for a car company detailed in [38]. The review also unveils a plethora of real-world applications in decision-making techniques, as seen in [39–41], etc. Furthermore, a significant contribution discussed within this literature is the algorithmic support for decision-making using fuzzy values, found in [42,43]. The exploration of defuzzification models for managing uncertainty in natural disaster scenarios, such as derailment incidents, is presented in [44]. Additionally, this review underscores the capability of fuzzy graph techniques to address complex challenges, like the partitioning of VLSI networks to reduce crossover. It delves into the strategy of achieving this through planar partition subgraphs

and deliberating on edge strength-based decisions. Finally, this comprehensive literature review provides an overview of past research on fuzzy graphs, offering valuable insights into their evolving role in decision-making.

The following section provides the methodology for finding the planar subgraphs by forming PP -subgraphs and fuzzy thickness values are discussed.

2.2. Methodology

In this paper the planar subgraph concepts are discussed in the environment of fuzzy graphs. Since different types of subgraphs are exist in both graph and fuzzy graph theory. But the formation of fuzzy planar subgraph is new and helps to modify the complex graph structures into simpler one to the greater extent. The following methods provides us to obtain the fuzzy planar subgraphs.

1. The formation of fuzzy planar subgraphs is obtained by the vertex deletion (VD) and edge deletion (ED) in the main fuzzy graph.
2. Generally the partition of fuzzy graph happened in the structure having intersecting edges. Additionally this can be done by choosing the maximum strength's fuzzy edge e such that the number of vertices and edges adjacent to the edge e forms a subgraphs of G and remaining other vertices and edges forms another subgraphs of G . This process is said to be Planar Partition Subgraph (PP -Subgraph).
3. One of the main property in planarity is Thickness. The fuzzy thickness value can be calculated by the help of PP -subgraph.

The following section provides the basics and important results for constructing the planar subgraphs in fuzzy graphs.

3. Preliminaries

The concept of subgraphs in classical graph theory is essential and has been extensively studied by researchers in last decades. Some enlisted works are mentioned here as Harary [3] introduced the notion of subsets in graphs, which are called subgraphs for a given graph G and provided with interesting properties. Lu [45] studied the results of deleting vertices and edges on induced graphs. Wilson [4] provided significant results related to the thickness of planar graphs. Further, some basic definitions are listed as follows.

In graph theory, a graph $H = (V', E')$ is considered a subgraph of $G = (V, E)$ if V' is a subset of V and E' is a subset of E . This relationship is denoted as $H \subseteq G$, which means that G is a supergraph of H . If $V' = V$, then H is called a spanning subgraph of G , as it spans across all the vertices of G . On the other hand, if $V' \subset V$, meaning V' is a proper subset of V , then H is called an induced subgraph of G . An induced subgraph is the maximal subgraph of G that can be obtained by deleting vertices from G .

Moreover, the subgraph and planar related definitions are presented below, serving as valuable steps towards constructing similar formations within the context of fuzzy graphs.

Definition 1 ([45]). A vertex deleted subgraph of a graph G is a subgraph $G - \{v\}$ obtained by deleting the vertex $v \in V$ from the graph G and all the edges incident to it.

Definition 2 ([45]). An edge deleted subgraph of a graph G is a subgraph $G - \{e\}$ obtained by deleting the edge $e \in E$ from the graph G .

Definition 3 ([3]). A graph can be geometrically drawn in many ways such that the crossing of edges are not involved in the plane surface then it is said to be 'planar graph'.

Definition 4 ([3]). A planar representation of a graph divides the plane into a number of connected regions called faces, each region bounded by the edges of the graph. The portion lying outside the graph in the plane is called infinite region.

Definition 5 ([3]). The thickness number is defined as the minimum number of planar subgraphs from G whose union is G .

Now, Let us examine the importance and basics of fuzzy graphs. The following definitions will provide a better understanding of how fuzzy planar graphs are constructed. In addition, the dual graph concept is depicted with example. The purpose of this is to understand the core concept of the paper that were discussed in detail.

Definition 6 ([5]). A fuzzy graph G is a triplet (V, σ, μ) , since the vertex set V is non void, the membership value of vertices are defined by $\sigma : V \rightarrow [0, 1]$ and the membership value of each edge is defined by $\mu : V \times V \rightarrow [0, 1]$ the σ and μ is a pair of functions in G , and satisfies the following condition: $\mu(x, y) \leq \min\{\sigma(x), \sigma(y)\} \forall x, y \in V$.

Definition 7 ([24]). Strength of the fuzzy edge $\mu(a, b)$ in G is measured by

$$I_{(a,b)} = \frac{\mu(a, b)}{\min(\sigma(a), \sigma(b))}.$$

If $I_{(a,b)} \geq 0.5$, the fuzzy edges are strong otherwise weak.

Definition 8 ([24]). Fuzzy graphs drawn on the plane surface such that no intersection of edges occurs, then the fuzzy graph is said to be fuzzy planar graph. Suppose (a, b) and (c, d) be the fuzzy edges in G , which are crossed together and p be the point of intersection of (a, b) and (c, d) . The intersecting value of p is calculated as

$$I_p = \frac{I_{(a,b)} + I_{(c,d)}}{2}$$

where $I_{(a,b)}$ and $I_{(c,d)}$ be the strength of the two fuzzy edges.

Definition 9 ([24]). Fuzzy planarity value in fuzzy graphs describes the amount of planarity of a fuzzy planar graph. Fuzzy planarity value is denoted by f and

$$f = \frac{1}{1 + \{I_{p_1} + I_{p_2} + 5 + I_{p_n}\}} = \frac{1}{1 + \sum_{i=1}^n I_{p_i}}.$$

If the number of the point of intersections in fuzzy graph increases, planarity decreases. Hence, I_p is inversely proportional to the planarity. Thus f is bounded and lies in $0 < f \leq 1$.

Definition 10 ([24]). When the fuzzy planarity value is 1, any fuzzy edges in G do not cross each other and thus each region is formed by the cycle, which constitute a fuzzy face in the fuzzy planar graph. The membership value of the fuzzy face is defined as $\min\left\{\frac{\mu(x,y)}{\min(\sigma(x), \sigma(y))} : \mu(x, y) \in E'_i\right\}$ where E'_i is the region bounded by the fuzzy edges in fuzzy planar graph.

Definition 11 ([24]). Let $G = (V, \sigma, \mu)$ be a fuzzy planar graph with the planarity value 1 and $E = \{\mu(x, y) : (x, y) \in V \times V\}$. Again, let $F_1, F_2, \dots, F_k$ be the fuzzy faces of G . The fuzzy dual graph of G is a fuzzy planar graph $G' = (V', \sigma', \mu')$, where $V' = \{x_i : i = 1, 2, \dots, k\}$ which is corresponding to the face F_i of G ; E' is the set of edges of G' , and $(x, y) \in E'$ if there is a common boundary for the faces F_i and F_j . The function σ' is given as follows: $\sigma' : V' \rightarrow [0, 1]$ such that $\sigma'(x)$ is the maximum among all the membership values of the edges of the boundary of F_i . Between two faces F_i and F_j of G , there may exist more than one common edge. Thus, between two vertices x and y in fuzzy dual graph G' , there may be more than one edges. If $(a, b) \in E$ is a common edge between the fuzzy faces F_i and F_j , then the membership value of the fuzzy edge $(x, y) \in V'$ is equal to the membership value of the edge (a, b) .

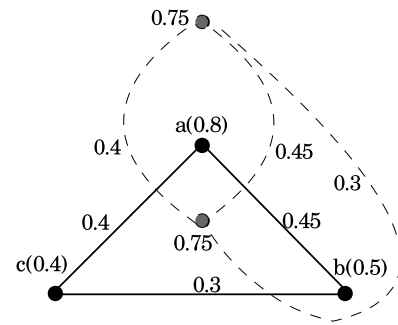


Fig. 1. Simple fuzzy graph G and superimposed fuzzy dual graph G' .

For instance, consider a fuzzy graph G as shown in Fig. 1 and the superimposed image of fuzzy dual graph G' presented in same figure.

Now, focussing on the formation of subgraphs from vertex and edge deletions in the following sections.

4. V D-fuzzy planar subgraph

We consider the fuzzy planar graph G with intersection of fuzzy edges such that $f \neq 1$.

Definition 12 (Vertex Deletion Fuzzy Subgraph or V D-fuzzy Subgraph). Let G be the fuzzy graph and $W \subseteq V$. The fuzzy subgraph obtained by deleting the set W from G and is denoted by $G - W$.

$$G - W = \{v \in V \setminus W : \mu(x, y) \neq 0, \forall x, y \in V \setminus W\}$$

The fuzzy graph $G - W$ is the V D-fuzzy subgraph of G

Definition 13. Let G be the fuzzy planar graph, If G' is the V D-fuzzy subgraph of G with planarity value $f_{G'} = 1$, then the subgraph G' is the V D-fuzzy planar subgraph of G .

Example 1. Consider the fuzzy graph $G = (V, \sigma, \mu)$ which is shown in Fig. 2(a). The vertices in G , $V = \{a, b, c, d, e, f\}$ and the membership value of vertices are $\sigma(a) = \sigma(f) = 0.5$, $\sigma(b) = 0.75$, $\sigma(c) = \sigma(d) = 0.3$, $\sigma(e) = 0.6$, and for edges $\mu(a, b) = 0.55$, $\mu(a, c) = \mu(c, f) = 0.25$, $\mu(b, c) = 0.15$, $\mu(b, e) = 0.5$, $\mu(b, f) = \mu(e, f) = 0.45$, $\mu(c, d) = \mu(c, e) = 0.3$ and $\mu(d, e) = 0.2$. There exist intersection of fuzzy edges between $\mu(b, f)$ and $\mu(d, e)$ in G . Let X and Y be the subsets in V and it contains $X = \{a(0.5)\}$ and $Y = \{d(0.3)\}$ of the fuzzy graph G .

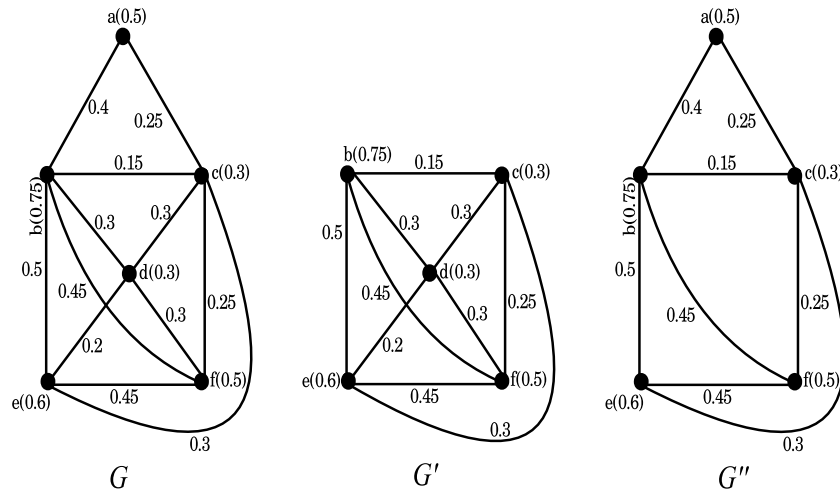
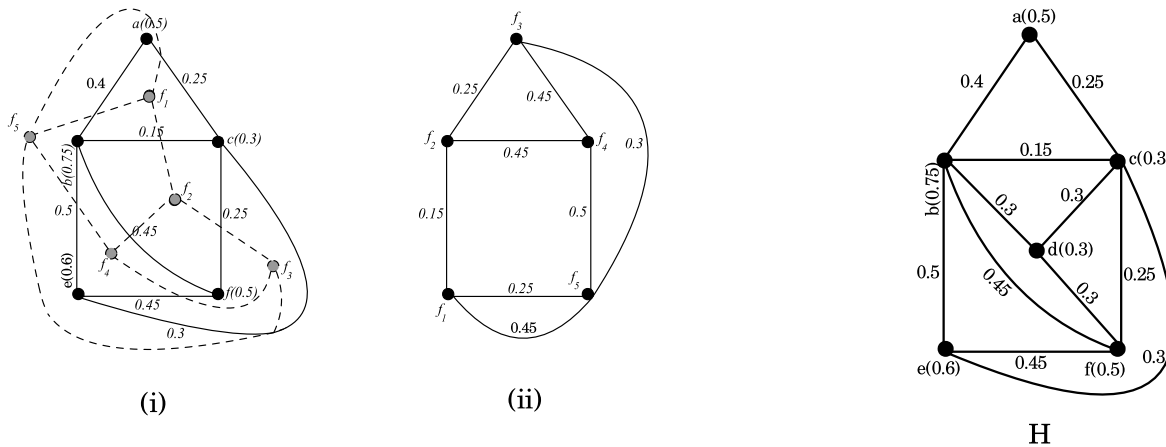
The fuzzy subgraphs $G - X = V \setminus X = \{v \in V \setminus X : \mu(x, y) \neq 0, x, y \in V \setminus X\} = G'$ and $G - Y = G''$ are shown in Fig. 2(b) and (c) respectively. Thus, the subgraph G' is a V D-fuzzy subgraph since $f_{G'} \neq 1$ whereas G'' is V D-fuzzy planar subgraph of G .

Theorem 1. Every V D-fuzzy subgraph in fuzzy planar graph G with $f_G = 1$, is always a V D-fuzzy planar subgraphs of G .

Proof. Let G be the fuzzy planar graphs with $f_G = 1$ and H be the V D-fuzzy subgraph of G . Since the fuzzy graph G has planarity value $f_G = 1$, so each subgraph in G do not have any intersections of edges. Thus, any vertex deleted subgraphs in G is a V D-fuzzy planar subgraph of G .

Note 1. The V D-fuzzy subgraph of G is always need not be the V D-fuzzy planar subgraph of G .

Theorem 2. Let G be connected fuzzy planar graph with $f_G \neq 1$ and $W \subseteq V$. If G' is the V D-fuzzy planar subgraph of G then it has the fuzzy dual graph.

Fig. 2. (a). fuzzy graph (b). VD -fuzzy subgraph & (c). VD -fuzzy planar subgraph.Fig. 3. Example for VD -fuzzy planar subgraph has fuzzy dual graphs.Fig. 4. Example for fuzzy dual graphs but not a VD -fuzzy planar subgraph.

Proof. Let G be a connected fuzzy planar graph with $f_G \neq 1$ and $W \subseteq V$. Let $G - W$ forms a VD -fuzzy planar subgraph G' with $f_{G'} = 1$. Since, there is no intersection of fuzzy edges in the subgraph G' would have finite number of regions. Each region is enclosed by the set of membership value of edges and is said to be fuzzy face value, defined by $F_i = \min\{\frac{\mu(x,y)}{\min\{\sigma(x),\sigma(y)\}} : \text{Each region enclosed by membership value of edges}\}$. Now, G'' be the fuzzy dual graph and let $v_1'', v_2'', \dots, v_n''$ be the vertices placed in every region such that the membership values of the vertices having the corresponding fuzzy face values $F_1, F_2, \dots, F_n$ respectively. Between two faces F_i and F_j of G' , there may exist more than one common edge. Thus, between two vertices v_i and v_j in fuzzy dual graph G'' there may be more than one edges. Because, if $\mu'(a, b)$ is a common edge between the fuzzy faces F_i and F_j , then the membership value of the fuzzy edge $\mu'(x, y)$ is equal to the membership value of the edge $\mu''(a, b)$ in G'' . Therefore, it satisfies as definition of fuzzy graph by the functions $\sigma'' : V'' \rightarrow [0, 1]$ and $\mu'' : V \times V \rightarrow [0, 1]$. Thus, the G'' is the fuzzy dual graph G' whereas G' is a VD -fuzzy planar subgraphs of G .

Example 2. By Example 1, the subgraph G'' is the VD -fuzzy planar subgraph of G and the dotted lines be superimposed dual graph of G'' shown in Fig. 3(i). The separate image of the fuzzy dual graph is shown in Fig. 3(ii). The fuzzy dual graph has fuzzy face values and the membership value of the fuzzy faces are $f_1 = f_2 = f_3 = f_5 = 0.5$ and $f_4 = 0.833$. Hence the VD -fuzzy planar subgraph G'' has a fuzzy dual graph.

Theorem 3. Fuzzy dual graph exists for some fuzzy subgraph in G and its always need not be the VD -fuzzy planar subgraph of G .

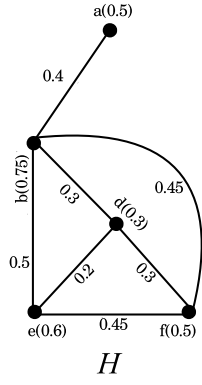
Proof. The proof is straightforward from the following illustration. Consider Fig. 2(a), the subgraph H obtained from G is shown in Fig. 4 and it becomes a fuzzy dual graph of G due to the their fuzzy face values for the each region in H . But H is not the VD -fuzzy planar subgraph of G because the deletion of the fuzzy edge $\mu(e, d)$ is not equal to any of the VD -fuzzy planar subgraphs. Hence, it completes the proof.

4.1. Maximum and maximal VD -fuzzy planar subgraphs

Here, the following discusses the maximum and maximal fuzzy planar subgraphs in terms of VD .

Definition 14. If $G' = (V', \sigma', \mu')$ is the VD -fuzzy planar subgraph of G with order (m) and size (n) such that there is no other VD -fuzzy planar subgraphs of G with maximum order and size compared to the subgraph G' . Then G' is said to be the maximum VD -fuzzy planar subgraph of G .

(i.e.) The order and size of the VD -fuzzy planar subgraphs G' and G'' of G are (m', n') and (m'', n'') respectively. The subgraph G' is said to be maximum VD -fuzzy planar subgraph of G if $(m', n') > (m'', n'')$.

Fig. 5. Example for Maximum VD -fuzzy planar subgraph.

Note 2. Suppose two VD -fuzzy planar subgraphs G_1 and G_2 from the fuzzy graph G . If $G_1 \neq G_2$ but the order and size of two subgraph are equal (i.e.), $(m', n') = (m'', n'')$, then the subgraphs are maximum VD -fuzzy planar subgraphs of G .

Example 3. Consider the fuzzy graph G from Fig. 2(a). The subset $W = \{c(0.3)\}$ is deleted from G , $G - W$ forms the subgraph H such that which is the highest maximum VD -fuzzy planar subgraph in G as shown in Fig. 5.

Theorem 4. Maximum VD -fuzzy planar subgraph has fuzzy dual graph, but the converse need not be true.

Proof. From the proofs of Theorems 2 & 3, it can be conclude that the maximum VD -fuzzy planar subgraph has fuzzy dual and converse need not to be true.

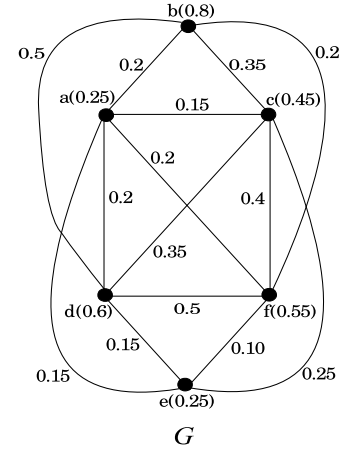
Definition 15. A VD -fuzzy planar subgraph $G' = (V', \sigma', \mu')$ induced on G and every other subgraphs of G induced by the vertex set $V'' = V' \cup \{v\}$ (where $v \in V \setminus V'$) has $f_{G' \cup \{v\}} \neq 1$, then G' is the maximal VD -fuzzy planar subgraph of G .

Theorem 5. Every maximum VD -fuzzy planar subgraph is maximal VD -fuzzy planar subgraph of G .

Proof. Let G' be the maximum VD -fuzzy planar subgraph of G . There exist a subset $W \neq \emptyset \subseteq V$ which contain some deleted vertices of G . Choose a vertex u in W , and consider $G'' = G' \cup \{u\}$ where $G'' = \{x \in V' \cup \{u\} : \mu(x, y) \neq \emptyset, x, y \in V' \cup \{u\}\}$. Since G' is maximum induced planar subgraph, if adding anyone of the vertex u will results in $f \neq 1$. Thus, G' is the maximal induced VD -fuzzy planar subgraph of G . Hence the results follows from the fact that every maximum induced VD -fuzzy planar graph is a maximal induced VD -fuzzy planar subgraph of G .

Theorem 6. The converse of Theorem 5 is always need be not true.

Proof. Let G be the fuzzy planar graph with $f_G \neq 1$ and, G' and G'' are the VD -fuzzy planar subgraphs of G . Let W_1, W_2 be the subsets of vertex set V in G . W_1 and W_2 contain the deleted vertices from G respectively. Then G' and G'' are said to be maximal fuzzy planar subgraphs if $G' \cup \{u\}$ where $u \in W_1$ and $G'' \cup \{v\}$ where $v \in W_2$ are non-planar. Therefore, these subgraphs are maximal VD -fuzzy planar subgraphs of G . However, the fuzzy subgraphs have different order and size, therefore the VD -fuzzy planar subgraph with highest order and size will be the maximum VD -fuzzy planar subgraph. Hence either G' or G'' will be the maximum VD -fuzzy planar subgraph, but still

Fig. 6. Example for maximal VD -fuzzy graph.

the two fuzzy graphs will be the maximal. Thus, every maximal VD -fuzzy planar subgraph need not be the maximum VD -fuzzy planar subgraph.

Note 3. Two VD -fuzzy planar subgraphs are maximal, only when the two subgraphs are maximum VD -fuzzy planar subgraphs by the Note 2.

Example 4. Consider the fuzzy planar graph G with planarity value $f = 0.4074$ as shown in Fig. 6. The membership value of the vertices are $\sigma(a) = \sigma(e) = 0.25$, $\sigma(b) = 0.8$, $\sigma(c) = 0.4$, $\sigma(d) = 0.6$ and $\sigma(f) = 0.5$. The membership values of edges are $\mu(a, b) = \mu(a, c) = \mu(a, f) = \mu(a, d) = \mu(b, f) = 0.2$, $\mu(a, e) = \mu(d, e) = 0.15$, $\mu(b, c) = \mu(c, d) = 0.35$, $\mu(c, e) = 0.25$, $\mu(c, f) = 0.4$ and $\mu(d, f) = 0.5$. Since G has intersection of fuzzy edges which is clearly observed from Fig. 6. The number of maximal VD -fuzzy planar subgraphs from the fuzzy graph G is shown in Table 1. If one more vertex is added to the subgraph then its planarity value $\neq 1$. The subset W_1 deleted from the fuzzy graph G (i.e., $G - W_1$), has the highest order and size compare to the other subgraphs in G . Hence $G - W_1$ is the maximum VD -fuzzy planar subgraph of G .

5. ED-fuzzy planar subgraph

Consider the fuzzy planar graph G with intersection of fuzzy edges such that $f \neq 1$. The subgraph obtained by deleting some of the edges in the fuzzy graph is said to be Edge-deletion (ED) subgraph. The following section provides the formal definition of ED -fuzzy planar subgraph and the results are presented.

Definition 16 (Edge Deletion Fuzzy Subgraph or ED -fuzzy Subgraph). Let $G = (V, \sigma, \mu)$ be the fuzzy graph and $E = \{\mu(x, y) : \mu(x, y) \neq 0, \forall x, y \in V\}$. Let U be the subset of E such that the deletion of the set U from E forms a subgraph called an ED -fuzzy subgraph of G . $G - U = \{\mu(x, y) \in E \setminus U : \mu(x, y) \neq 0, \forall x, y \in V\}$. Then, the fuzzy graph $G - U$ is an ED -fuzzy subgraph of G .

Definition 17. Let G be the fuzzy planar graph and G' be an ED -fuzzy subgraph of G with $f_{G'} = 1$, then the subgraph G' is an ED -fuzzy planar subgraph of G .

Theorem 7. Every ED -fuzzy subgraph in fuzzy planar graph G whose planarity value is $f_G = 1$ then it is always an ED -fuzzy planar subgraph of G .

Proof. Let G be the fuzzy planar graphs with $f_G = 1$ and H be an ED -fuzzy subgraph of G . Since the fuzzy graph has planarity value $f_G = 1$,

Table 1
Number of maximal VD -fuzzy planar subgraphs.

S. No.	Maximal VD -fuzzy planar subgraphs	subset of V	Order and size
1.		$W_1 = \{e(0.25)\}$	(2.65, 2.55)
2.		$W_2 = \{a(0.25)\}$	(2.55, 2.3)
3.		$W_3 = \{c(0.25)\}$	(2.4, 1.5)
4.		$W_4 = \{d(0.6)\}$	(2.2, 2)
5.		$W_5 = \{f(0.5)\}$	(2.3, 1.5)

so each subgraph in G does not have any intersections. Thus, any edge deleted subgraphs in G is an ED -fuzzy planar subgraphs of G .

Note 4. An ED -fuzzy subgraph of G is always need not be an ED -fuzzy planar subgraph of G .

Remark 1. Let G be the connected fuzzy planar graph with $f_G \neq 1$ and $W \subseteq E$. If G' is an ED -fuzzy planar subgraph of G then it has the fuzzy dual graph.

5.1. Maximum and maximal ED -fuzzy planar subgraphs

Here, the following discusses the maximum and maximal fuzzy planar subgraphs in terms of ED .

Definition 18. Let $G' = (V, \sigma, \mu')$ be an ED -fuzzy planar subgraph of G with n' be the size of the graph G' and there is no other ED -fuzzy

planar subgraphs of G has greatest size compared to n' . Thus, G' is said to be the maximum ED -fuzzy planar subgraph of G .

Note 5. Suppose two ED -fuzzy planar subgraphs G_1 and G_2 from the fuzzy graph G . If $G_1 \neq G_2$ but the size of the two subgraphs are equal (i.e.) $n' = n''$, then the two subgraphs are maximum ED -fuzzy planar subgraphs of G .

Definition 19. A fuzzy planar subgraph G' on G and every other subgraphs of G induced by the edge set $E'' = E' \cup \{\mu(x, y)\}$ with $\mu(x, y) \in E \setminus E'$ has planarity value $0 < f < 1$, then G' is called the maximal fuzzy planar subgraph of G .

Remark 2. Every maximum ED -fuzzy planar subgraph in G is maximal fuzzy planar subgraph of G and the converse is need not be true.

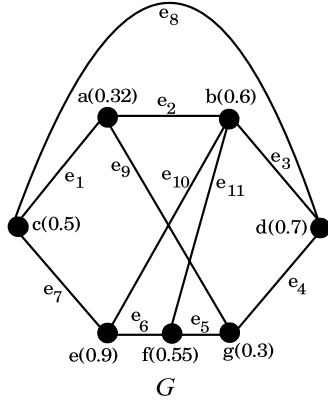
Fig. 7. Example for maximum and maximal ED -fuzzy subgraphs from fuzzy graph G .

Table 2

Number of maximal ED -fuzzy planar subgraphs.

S. No.	Maximal ED -fuzzy planar subgraphs	Subset of V	Order and size
1.		$U_1 = \{e_{10}, e_{11}\}$	(2.95)
2.		$U_2 = \{e_9\}$	(3.6)

Example 5. Consider the fuzzy planar graph G as shown in Fig. 7 has planarity value $f = 0.36738$. The vertex set $\sigma(a) = 0.32$, $\sigma(b) = 0.6$, $\sigma(c) = 0.5$, $\sigma(d) = 0.7$, $\sigma(e) = 0.9$ and $\sigma(f) = 0.55$. For edges $\mu(a, b) = e_i$, $\forall \mu(a, b) \in E, i = 1, 2, \dots, 11$, and edge set as $\mu(e_1) = \mu(e_8) = \mu(e_9) = 0.3$, $\mu(e_2) = 0.25$, $\mu(e_3) = 0.45$, $\mu(e_4) = 0.15$, $\mu(e_5) = 0.2$, $\mu(e_6) = \mu(e_7) = 0.5$, $\mu(e_{10}) = 0.55$ and $\mu(e_{11}) = 0.4$. The maximal ED -fuzzy planar subgraphs are shown in Table 2. Thus, the two subgraphs are maximal and $G - U_2$ is the maximum ED -fuzzy planar subgraph of G .

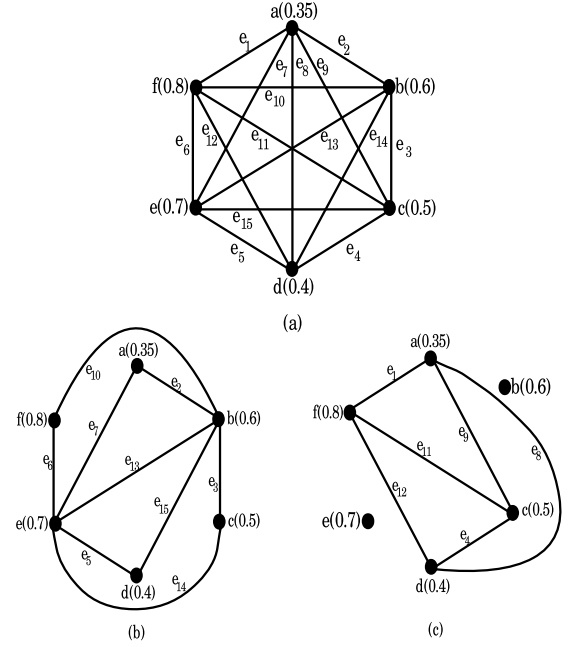
Note 6. Two ED -fuzzy planar subgraphs are maximal, only when the two subgraphs are maximum ED -fuzzy planar subgraphs by the Note 5.

6. Planar partition subgraphs (PP-subgraphs)

This section explains the procedure for partitioning the main fuzzy graph when $f \neq 1$.

6.1. PP-subgraphs

Let G be the fuzzy graph with $f_G \neq 1$ and again let $P_1, P_2, P_3, \dots, P_n$ be the point of intersections of fuzzy edges in the main graph G . If $I_{(a,b)}$ denotes the strength of fuzzy edge in P_i . Choose a fuzzy edge with maximum strength in P_i such that the membership value of the vertices and edges which are incident to the maximum strength fuzzy edge forms a subgraph G' of G . This will be the first partitioned subgraph of G . The next subgraph is obtained by $G'' = G - G'$ with $0 < f \leq 1$. If the partitioned subgraphs have planarity value to be 1, then the processes stops. Otherwise, the subgraph G'' is partitioned by

Fig. 8. Example for PP -subgraphs (a). Fuzzy graph $G = K_6$ (b) & (c) Fuzzy planar subgraphs G' and G'' respectively.

choosing another maximum strength's fuzzy edge in the intersections and forming the subgraph by the number of vertices and edges adjacent to this particular fuzzy edge. Likewise, the subgraphs are partitioned in the fuzzy graph G .

Example 6. Consider the fuzzy graph $G = K_6$ as shown in Fig. 8. The fuzzy graph $G = K_6$ whose vertex set $\sigma(a) = \{0.35$, $\sigma(b) = 0.6$, $\sigma(c) = 0.5$, $\sigma(d) = 0.4$, $\sigma(e) = 0.7$ and $\sigma(f) = 0.8\}$ and edge set $E = \{\mu(e_1) = \mu(e_5) = \mu(e_7) = \mu(e_{11}) = \mu(e_{15}) = 0.3$, $\mu(e_2) = \mu(e_8) = 0.25$, $\mu(e_3) = \mu(e_4) = \mu(e_{14}) = 0.4$, $\mu(e_6) = \mu(e_9) = \mu(e_{12}) = 0.2$, $\mu(e_{10}) = 0.5$ and $\mu(e_{13}) = 0.6\}$. Choosing the maximum strength of the fuzzy edge $I_{(b,e)} = \frac{0.6}{\min\{0.6, 0.7\}} = 1$ and the corresponding fuzzy edge is $\mu(b, e)$ and forming a subgraph by the vertices and edges which are adjacent to the edge $\mu(b, e)$ and named as G' whose planarity value $f_{G'} = 1$ as shown in Fig. 8(b). Therefore, $G - G' = G''$ whose planarity value is also $f_{G''} = 1$ as shown in Fig. 8(c). Hence, G' and G'' are the PP -subgraphs of G .

6.2. Algorithm for finding PP -subgraphs of G

This section provides the algorithm for PP -subgraphs from the fuzzy graph G .

Require: $G = (V, \sigma, \mu)$

Ensure: $G', G'', \dots, G^n$ are the PP -subgraphs of G

- 1: Compute the crossing edges in the fuzzy graph G
- 2: Let $P_1, P_2, \dots, P_n$ be the point of intersections in G .
- 3: Note the membership values of the edges in $P_1, P_2, \dots, P_n$.
- 4: To find the strength of the intersecting edges at P_i by $I_{(v_i, v_j)} = \frac{\mu(v_i, v_j)}{\min\{\sigma(v_i), \sigma(v_j)\}}$
- 5: $K = \max\{I_{(v_i, v_j)} : (v_i, v_j) \in G, i \neq j\}$, fix any one edge if two edges have same strength values.
- 6: Form the fuzzy subgraph by the number of vertices and edges incident with the K and set the subgraph as G' .
- 7: To find the another subgraph by $G - G' = G''$.
- 8: **if** $f_{G'} = 1 = f_{G''}$ **then**
- 9: Stop
- 10: **else**
- 11: Follow the Steps from 1–8, form the subgraphs $G''', \dots, G^n$

12: **end if**

13: The algorithm stops when all the fuzzy planar subgraphs with the planarity value $f_{G'} = f_{G''} = 5 = f_{G^n} = 1$.

6.2.1. Performing the computation for Example 6

Consider the fuzzy graph $G = K_6$ with the membership values of vertices and edges.

Step 1: To compute the crossing of the edges in G .

Step 2: Name the point of intersection of G as $P_1, P_2, \dots, P_{13}$.

Step 3: Collect the membership values of the edges in G especially the intersecting edge's membership values.

Step 4: The strength of the intersecting edges are $I_{e_7} = 0.8571$, $I_{e_8} = 0.7142$, $I_{e_9} = 0.05714$, $I_{e_{10}} = 0.8334$, $I_{e_{11}} = I_{e_{15}} = 0.6$, $I_{e_{12}} = 0.5$, $I_{e_{13}} = I_{e_{14}} = 1$.

Step 5: $K = \{I_{e_{13}}, I_{e_{14}}\}$, fix the fuzzy edge $K = I_{e_{13}}$.

Step 6: Form the subgraph G' , the number vertices and edges which are incident to K .

Step 7: $G'' = G - G'$ is also the fuzzy planar subgraph of G .

Step 8: If G' and G'' has $f_{G'} = f_{G''} = 1$, then the fuzzy planar subgraphs are reached.

By following the same procedures can partition any graph G with finite number of fuzzy planar subgraphs.

Theorem 8. Let $G', G'', \dots, G^n$ are the PP -subgraphs of G . Then

- (i). $\bigcup_{i=1}^n G^i = G$
- (ii). $\bigcap_{i=1}^n G^i = U \subseteq V$

Proof. The two cases are obviously true by the definition of PP -subgraphs.

Example 7. Consider the fuzzy graph $G = K_6$ and let G' and G'' are fuzzy planar subgraphs formed by the PP -subgraphs with fuzzy edge $\mu(e_{14})$. If the union of the fuzzy subgraphs gives the fuzzy graph $G = K_6$ ((i.e.) $G' \cup G'' = G$), where the intersection of the fuzzy planar subgraphs $G' \cap G'' = U$. U contains all the vertices of V but no edges in common.

Limitations: Let us now discuss some limitations of this model.

1. The PP -subgraphs hold good for fuzzy graphs with planarity value less than 1. But if $f = 1$, then PP -subgraph not applied for the graph due to the non-intersection of edges.
2. This model enables the partitioning of the graph into planar subgraphs in both in a crisp and fuzzy graphs. In the crisp sense, all edges and vertices possess an equal membership value as 1, allowing to partition the graph by choosing any edge due to the uniform strength of edges whereas in fuzzy graph it focuses on higher strength to partition and form the planar subgraph.

Definition 20. Let G be the fuzzy planar graph with $f_G \neq 1$, there exist finite number of planar subgraph by PP -subgraph using the edge I_e on G such that the total number of planar subgraphs formed by PP -subgraph is denoted by ρ .

Suppose the fuzzy planar graph has finite number of intersection of edges in G . Consider the two point of intersections have the same maximum strength. Then, the number of fuzzy planar subgraphs formed by PP -subgraph by two different edges of G are ρ_i and ρ_j may or may not equal $\forall i, j \in N$.

Note 7. If the point of intersection in G is only one, then $\rho = 2$.

For instance, consider the fuzzy graph $G = K_5$ has the point of intersection at one point only (i.e.) I_{P_1} . Thus, the total number of subgraphs formed by PP -subgraphs is 2 (i. (e) $\rho = 2$).

Theorem 9. If the maximum strength of the fuzzy edges $I_{(a,b)}$ and $I_{(c,d)}$ are equals, but the subgraphs formed with respect to $I_{(a,b)}$ and $I_{(c,d)}$ are distinct.

Proof. The theorem is proved by the following counter examples.

Consider Fig. 8(a), the membership value of the $\mu(e_{14}) = 0.4$ and the strength of fuzzy edge be $I_{e_{14}} = \frac{\mu(e_{14})}{\min\{\sigma(b), \sigma(d)\}} = \frac{0.4}{0.4} = 1$. Therefore, $I_{e_{14}} = I_{e_{13}} = 1$. The PP -subgraphs formed by $I_{e_{14}}$ is $E = \{G^1, G^2\}$ as shown in Fig. 9. Hence, the subgraph G^1 have $f_{G^1} = 1$ and subgraph $G^2 = G - G^1$ have the planarity value as 1. The number of subgraphs of formed by PP -subgraph by the strength of the edges $I_{e_{14}}$ and $I_{e_{13}}$ are $\rho_1 = \rho_2 = 2$ are respectively. Hence, the subgraphs formed with respect to $I_{e_{13}}$ is not same as the subgraphs formed with respect to $I_{e_{14}}$. Thus, it completes the proof.

Definition 21. Let $\rho_1, \rho_2, \dots, \rho_n$ be the total number of subgraphs formed by PP -subgraphs using different fuzzy edges of fuzzy graph G , then

$$\rho^* = \min\{\rho_i, \forall i = 1, 2, \dots, n\}$$

where ρ^* is the minimum number of subgraphs formed by PP -subgraphs.

Example 8. By Example 6, the strength of the intersecting fuzzy edges $I_{e_{13}}$ and $I_{e_{14}}$ are same with $\rho_1 = \rho_2 = 2$. Therefore, $\rho^* = \min\{\rho_1, \rho_2\} = 2$.

7. Thickness value of fuzzy planar graphs

Graph Thickness or Thickness number is one of the identities of a planar graph. The thickness number is obtained by dividing the non-planar graphs into the finite number of planar subgraphs such that the union of these planar subgraphs is again G . The minimum number of planar subgraphs formed in G is the thickness number of graph G . The planar graph without crossing edges and the thickness number of the graph is 1. When the crossing of edges increases the thickness number also increases. Now, the notion is constructed in the fuzzy planar graph which is different from the graph approach.

Definition 22. Let G be the connected fuzzy planar graph with $0 < f < 1$. The thickness value of G is denoted by $Th(G)$ and defined as

$$Th(G) = \frac{I_{P_1} + I_{P_2} + 5 + I_{P_n}}{n + \rho^*}$$

where I_{P_i} , $n = 0, 1, 2, \dots, n$

Note 8. Fuzzy planar graph with $f = 1$, then $Th(G) = 0$.

Note 9. As the number of intersection of fuzzy edges in G and ρ^* increases, then the thickness value $Th(G)$ decreases. Further the thickness value of the fuzzy planar graph lies in the interval $0 \leq Th(G) < 1$.

Example 9. Consider the fuzzy graph $G = K_5$ is shown in Fig. 10. The membership values of the vertices and edges be $\sigma(a) = \sigma(d) = \sigma(e) = 0.5$, $\sigma(b) = 0.4$ and $\sigma(c) = 0.6$, and $\mu(a, b) = \mu(c, d) = 0.1$, $\mu(a, c) = \mu(a, d) = 0.5$, $\mu(b, d) = \mu(b, e) = 0.4$, $\mu(a, e) = 0.3$ and $\mu(c, e) = 0.25$. Therefore the thickness value of G is calculated as follows:

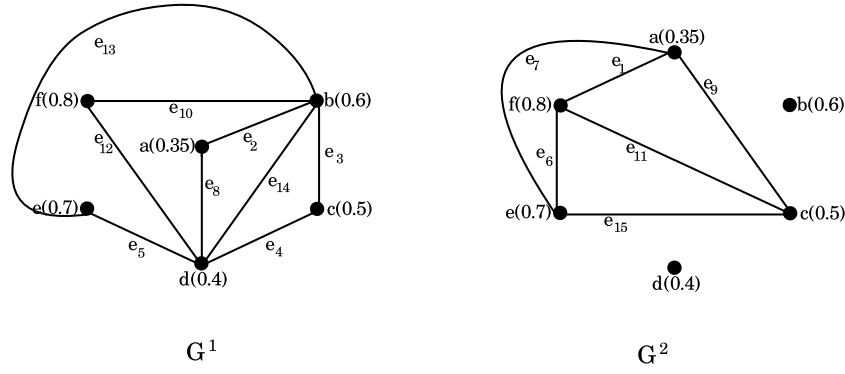
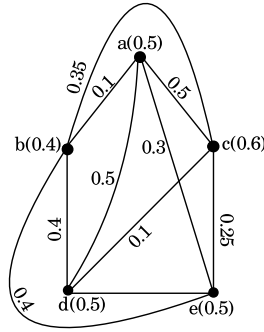
$$I_{(c,d)} = \frac{0.1}{\min\{0.5, 0.6\}} = 0.2 \text{ and } I_{(a,e)} = \frac{0.3}{\min\{0.5, 0.5\}} = 0.6$$

$$I_{P_1} = \frac{I_{(a,e)} + I_{(c,d)}}{2} = \frac{0.2 + 0.6}{2} = \frac{0.8}{2} = 0.4$$

$$Th(G) = \frac{I_{P_1}}{1 + \rho^*}$$

where ρ^* is the minimum number of fuzzy planar subgraphs in PP -subgraphs.

$$Th(G) = \frac{0.4}{1 + 2} = 0.1333$$

Fig. 9. Subgraphs formed by PP-subgraphs for the fuzzy edge $\mu(e_{14})$.Fig. 10. Fuzzy graph $G = K_5$.

Theorem 10. Let G be the fuzzy planar graph with $0 < f \leq 1$. The Thickness of fuzzy graph satisfies

$$Th(G) \leq 1 - f \quad (1)$$

Proof. Consider a fuzzy planar graph with planarity value as 1. Then the thickness of fuzzy graph is given by

$$Th(G) = \frac{I_{P_1} + I_{P_2} + 5 + I_{P_n}}{n + \rho^*} = \frac{0}{1} = 0$$

where $I_{P_n} = 0$, $n \in N$ and $\rho^* = 1$ where G is a subgraph of itself. By Eq. (1),

$$Th(G) = 0$$

Since $f = 1$, then $Th(G) = 1 - 1 = 0$ Hence the result satisfied for the fuzzy planar graph $f = 1$.

Now, consider a fuzzy planar graph with planarity value $0 < f < 1$,

$$\begin{aligned} f &= \frac{1}{1 + I_{P_1} + I_{P_2} + 5 + I_{P_n}} \\ 1 - f &= 1 - \frac{1}{1 + I_{P_1} + I_{P_2} + 5 + I_{P_n}} \\ &= \frac{1 + I_{P_1} + I_{P_2} + 5 + I_{P_n} - 1}{1 + I_{P_1} + I_{P_2} + 5 + I_{P_n}} \\ &= \frac{I_{P_1} + I_{P_2} + 5 + I_{P_n}}{1 + I_{P_1} + I_{P_2} + 5 + I_{P_n}} \end{aligned}$$

Consider $n + \rho^* \geq 1 + I_{P_1} + I_{P_2} + 5 + I_{P_n}$, where each I_{P_i} value lies in the interval $(0, 1]$ and n be the total number of point of intersections and ρ^* be the minimum of the total number of subgraph by the PP-subgraphs in G .

$$Th(G) = \frac{I_{P_1} + I_{P_2} + 5 + I_{P_n}}{n + \rho^*} \leq \frac{I_{P_1} + I_{P_2} + 5 + I_{P_n}}{1 + I_{P_1} + I_{P_2} + 5 + I_{P_n}} = 1 - f$$

Hence, $Th(G) \leq 1 - f$.

Example 10. In Fig. 10, the membership value of vertices and edges of fuzzy planar graph G assumed to be 1. The thickness value of fuzzy graph G is

$$I_{(c,d)} = 1 \text{ and } I_{(a,e)} = 1$$

$$I_{P_1} = \frac{I_{(a,e)} + I_{(c,d)}}{2} = \frac{1 + 1}{2} = \frac{2}{2} = 1$$

$$Th(G) = \frac{I_{P_1}}{1 + \rho^*} = \frac{1}{3} = 0.3334$$

Therefore,

$$1 - f = 1 - \frac{1}{1 + I_{P_1}} = 1 - \frac{1}{1 + 1} = 1 - 0.5 = 0.5$$

Thus, the theorem is true for the fuzzy graph has all the membership values are 1, (i.e), $Th(G) < 1 - f$.

Example 11. In Fig. 11, the membership value of the vertices be $\sigma(a) = \sigma(f) = 0.5$, $\sigma(c) = \sigma(d) = 0.3$, $\sigma(b) = 0.75$ and $\sigma(e) = 0.6$. The membership value of the edges be $\mu(a,d) = \mu(c,f) = 0.15$, $\mu(d,e) = \mu(b,c) = \mu(c,d) = 0.25$, $\mu(a,b) = 0.35$, $\mu(a,f) = 0.45$, $\mu(e,f) = 0.4$ and $\mu(b,e) = 0.6$. The thickness value is calculated as follows.

$$I_{(a,f)} = \frac{0.45}{\min\{0.5, 0.5\}} = 0.9, \quad I_{(b,e)} = \frac{0.6}{\min\{0.6, 0.75\}} = 1$$

$$I_{(c,d)} = \frac{0.25}{\min\{0.3, 0.3\}} = 0.8333$$

$$I_{P_1} = \frac{I_{(a,f)} + I_{(b,e)} + I_{(c,d)}}{3} = \frac{0.9 + 1 + 0.8333}{3} = 0.9111$$

Since $I_{(b,e)}$ is greater strength comparing with other edges the subgraph G' is formed by adjacent vertices and edges to $\mu(b,e)$ in G as shown Fig. 11(b). Subgraph G'' formed by $G'' = G - G'$ as shown in Fig. 11(c) such that $G' \cup G'' = G$. Therefore, $\rho^* = 2$

$$Th(G) = \frac{0.9111}{3} = 0.3037$$

$$1 - f = 1 - \frac{1}{1 + 0.9111} = 1 - \frac{1}{1.9111} = 1 - 0.52325 = 0.47675$$

Hence, $Th(G) < 1 - f$.

8. Application of PP-subgraph in VLSI design

A circuit or network is typically represented by graphs, especially in the context of VLSI network designs that make use of planarity. It is sometimes difficult to fix a single layer of a chip or a single board to a VLSI circuit during design. A circuit must be designed so that the number of layers, chips, board, and the delay in the circuit need to be minimized. Also, the number of interconnections and power consumption must also be considered when designing circuits. Generally, the circuit's vertices and edges have specific weights; these are

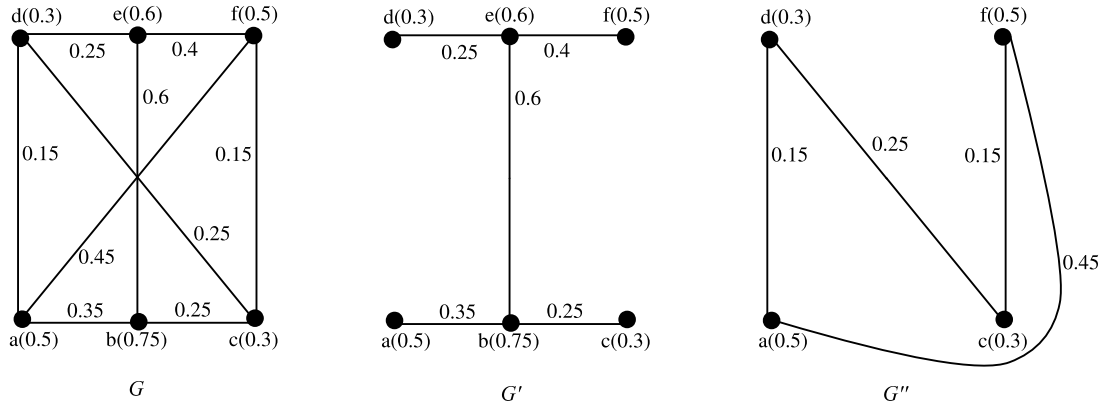
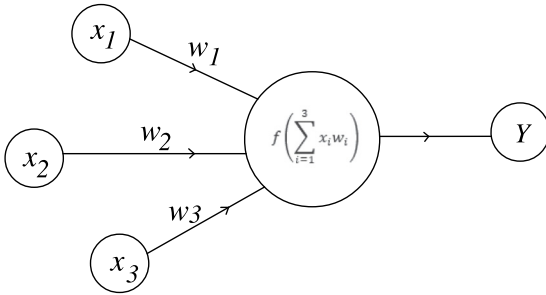
Fig. 11. (a). Fuzzy graph G (b). & (c). PP -subgraphs of G .

Fig. 12. Simple back propagation neural network.

membership values of the circuit. When the circuit is too large and hard to fix on a single device, this can be partitioned [46,47] into a finite number of sub-circuits. Algorithms for partitioning the VLSI networks through GEA (Genetic algorithm approach) and CTR (Counter based graph centric method) are proposed in [48].

In VLSI design, the hardware components are vertices, and the connecting wires are edges. VLSI implementation on artificial neural networks are usually considered to be efficient for the small area and low power consumption. Some of the power estimation method for VLSI circuit using Back Propagation Neural Network ($BPNN$) were proposed in [49]. For instance, the membership values of the vertices and edges for the simple back propagation neural network model is shown in Fig. 12. Each membership value of vertex and edge lies in the interval $[0,1]$. Thus, $x_i = 1 \forall i = 1, 2, 3$ and $w_1 = 0.2, w_2 = 0.4, w_3 = 0.6$ respectively. Thus, $\sum_{i=1}^3 x_i w_i = 1.2$ and the output for the back propagation network is $Y = \frac{1}{(1+e^{-1.2})} = 0.7685$.

As a direct result of this research is a high-level abstract graph-partitioning model. This model allows one to specify mathematical

evaluation metrics and control parameters, permitting algorithms and allowing one to identify the particular greatest edge strength of the network and forming subcircuit based on the PP -subgraph algorithm. To minimize the delay in sub-circuits, a general delay model was developed by Murgai et al. [50] and states as follows,

1. Each vertex may have a certain unique delay.
2. No delay exists between vertices in the same subgraph.
3. There is a constant delay between different subgraphs.

CTR Method: There various algorithm methods are available to sub-divide the VLSI networks. Some of them are ECB, CBC, CFB, OFB, CTR, etc. are effectively applied in the VLSI networks. CTR (counter), is based on graph center metrics. This approach involves establishing a graph center, extracting a subgraph based on this center, and repeating this procedure until all vertices were assigned in a subcircuit. The ISCAS85 C-17 benchmark circuit from the VLSI designs [51] and it is Gate diagram and schematics diagram in term of fuzzy graph is shown Fig. 13(a & b).

In Fig. 13(b)'s network diagram, the presence of edge intersections suggests the applicability of the PP -subgraph concept. This concept facilitates breaking down the diagram into finite sub-circuits. Both vertices and edges possess specific fuzzy values as defined by [52], with assumed membership values of 1 in this context. Notably, if the analyzed network features an intersected edge such as e_8 , which holds greater strength than other edges in the circuit, the PP -subgraph method identifies the sub-circuit formed by vertices and edges connected to e_8 , while the remaining vertices and edges compose another sub-circuit. These sub-circuits qualify as fuzzy planar subgraphs with a membership value of $f = 1$. Represented by rectangular enclosures, these sub-circuits are interconnected by links, effectively reducing intersections as depicted in Fig. 14.

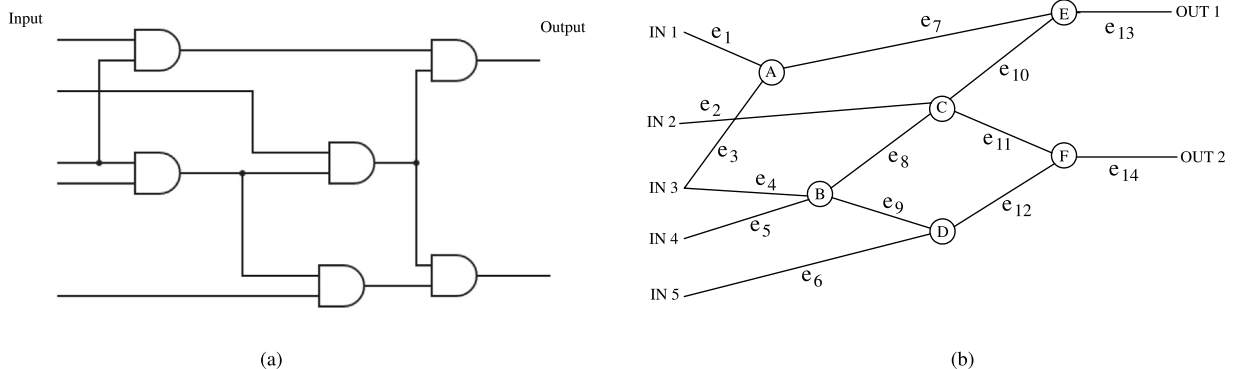


Fig. 13. (a). Gate diagram for ISCAS85 C-17 benchmark circuit (b). Schematic diagram of ISCAS85 C-17 benchmark circuit.

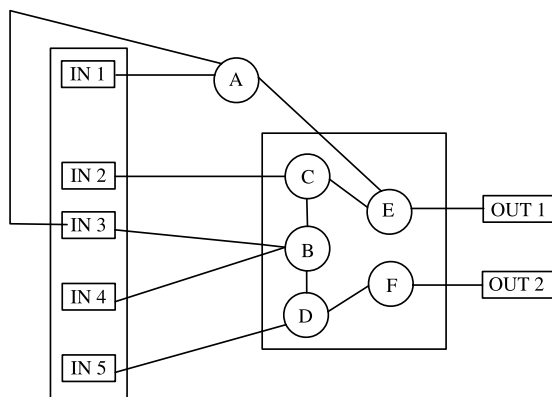


Fig. 14. Subcircuits network diagram of ISCAS85 C-17 benchmark circuit.

This theory assists in dividing complex structured networks into a finite set of sub-circuits. Connecting these sub-circuits minimizes intersections and mitigates issues related to timing complexities and intersection faults. Consequently, this approach holds promise for future applications.

9. Conclusion

This paper provides a comprehensive study on various aspects of planar subgraphs in fuzzy graphs. The main aim of the paper is to construct the planar partition subgraphs (*PP*-subgraph) and fuzzy thickness values in fuzzy graphs. This was achieved by the knowing the subgraph concepts, and some of the new types of subgraph operations such as *VD* (vertex deletion) and *ED* (edge deletion) results are presented. Further, the results and theorems on maximal and maximum planar subgraphs of *VD* and *ED* are provided with certain illustrations. The proposed methods and algorithms offer the efficient techniques for identifying the fuzzy planar subgraphs in fuzzy graphs, and the examples and illustrations provided help enhance the understanding of the concepts presented in this paper. Also, the notable work on fuzzy thickness value has certain important in the fuzzy graph approach as its one of the important planar property in graphs. The research contributes to the field of fuzzy graph theory by understanding the planar subgraphs in fuzzy graphs and providing new insights into their properties and applications. Moreover, these algorithms are expected to have practical implications in various real-world scenarios where fuzzy graphs are used to model uncertain or imprecise information. The paper also identifies potential avenues for further research in this area, which could inspire future studies and advancements in the field of fuzzy graph theory.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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